CASE D

Patient demographics

Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Reduced cardiac output (impaired EF)
- Bilateral peripheral oedema — reduced ankle mobility
- Orthopnoea — positional dyspnoea
- Skeletal muscle deconditioning / cardiac cachexia
- Severely reduced exercise tolerance
- Dyspnoea completing ADLs (showering, dressing)
- Reduced walking distance / community mobility
- Social isolation and reduced quality of life

■■ RED FLAGS

- requiring medical attention before physiotherapy exercise.)
- How far can you walk on flat ground before you have to stop, and how does that compare to 2 weeks ago?
- (Quantifies
- functional deterioration — allows me to establish pre-admission
- baseline and classify NYHA functional class, which directly guides my
- exercise prescription intensity and realistic goals.)
- Have you been weighing yourself daily? Have you noticed a sudden increase in weight recently?
- weight gain — 2 kg or more in 2–3 days — is the most sensitive early

■ PATHOPHYSIOLOGY (KEY POINTS)

- In HFrEF secondary to ischaemic heart disease , prior myocardial infarction has caused irreversible loss of contractile myocardium — replaced by non-compliant fibrous scar tissue.
- This reduces the left ventricle's stroke volume and ejection fraction (typically <40%), meaning the heart pumps inadequate blood forward to meet the body's metabolic demands despite normal or elevated venous return.
- While initially adaptive, chronic RAAS activation causes progressive fluid overload , increasing both preload and afterload on the already failing ventricle — perpetuating the cycle of deterioration.
- The elevated left ventricular end-diastolic pressure is transmitted retrogradely through the left atrium to the pulmonary veins, raising pulmonary capillary hydrostatic pressure .
- When this exceeds the plasma oncotic pressure (approximately 25 mmHg), fluid leaks from pulmonary capillaries into the interstitium and — in severe decompensation — into the alveolar spaces.
- This pulmonary oedema impairs gas diffusion, causes V/Q mismatch, reduces lung compliance, and substantially increases the work of breathing.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Resting Vital Signs + Orthostatic Blood Pressure Assessment

What it measures: Haemodynamic stability — determines whether it is safe to proceed with any physiotherapy intervention. This is the non-negotiable first step in every decompensated HF session.

baseline before any intervention.

Unsafe to exercise if: SpO■ <90% on supplemental O■, SBP <90 mmHg, HR >120 or <40 bpm, RR >30 bpm, new chest pain, or symptomatic orthostatic hypotension (SBP drop >20 mmHg with dizziness).

■ TREATMENT PLAN

- — is entirely dependent on haemodynamic stability. In decompensated
- HFrEF, this changes session to session and I cannot prescribe exercise
- safely without confirming it first.
- Expected Auscultation:
- late-inspiratory crackles — these reflect fluid in the alveoli and
- interstitium. Critically, these do NOT clear with coughing
- (distinguishing them from bronchiectasis crackles). Diminished air entry
- at bases if pleural effusions are present. Possible third heart sound
- (S3 gallop) — pathognomonic of volume overload in HFrEF. Possible wheeze
- only if concurrent obstructive airway disease.
- RR: 20–28 bpm (tachypnoeic)
- SpO■: 88–94% on O■